

PMD database

Table of the combined 100 most frequent PMDs detected in the 5 types of locations detected with the rda function of the PMD package (total of 118 PMD).

PMD	Reference	Putative transformation	Interpretation	Instrument	Study matrix	Reference type
0.036	1	-CH ₄ +O		FTICR-MS	environmental	Experimental study
	2	-CH ₄ +O		FTICR-MS	food	Experimental study
	3	-CH ₄ +O		FTICR-MS	environmental	Experimental study
	4	-CH ₄ +O		LC-MS/MS	living	Experimental study
	5,6	-CH ₄ +O		FTICR-MS	environmental	Curated database
1.979	1	-CH ₂ +O		FTICR-MS	environmental	Experimental study
	7	-CH ₂ +O		FTICR-MS	industrial	Experimental study
	5,6	-CH ₂ +O		FTICR-MS	environmental	Curated database
2.016	1	H ₂		FTICR-MS	environmental	Experimental study
	8	H ₂		HRMS	living	Methods study
	5,6,9	H ₂	hydrogenation/dehydrogenation	FTICR-MS	environmental	Curated database
2.052	1	-CH ₆ +O		FTICR-MS	environmental	Experimental study
	5,6	-CH ₆ +O		FTICR-MS	environmental	Curated database
3.959	<i>Not referenced</i>					
3.995	1	-C+O		FTICR-MS	environmental	Experimental study
	5	-C+O		FTICR-MS	environmental	Curated database
4.031	1	-H ₄		FTICR-MS	environmental	Experimental study
	5,6	-H ₄		FTICR-MS	environmental	Curated database
5.989	5	+C ₂ - H ₂ - O		FTICR-MS	environmental	Curated database

6.011	1	-C+H ₂ O		FTICR-MS	environmental	Experimental study
	5,6	-C+H ₂ O		FTICR-MS	environmental	Curated database
6.047	5,6	H ₆		FTICR-MS	environmental	Curated database
7.969	1	-H ₄ + C		FTICR-MS	environmental	Experimental study
	5,6	-H ₄ + C		FTICR-MS	environmental	Curated database
8.005	1	-C ₂ + O		FTICR-MS	environmental	Experimental study
	5,6	-C ₂ + O		FTICR-MS	environmental	Curated database
8.041	5,6	+C ₃ H ₄ - O ₂		FTICR-MS	environmental	Curated database
9.948	1	-H ₆ + O		FTICR-MS	environmental	Experimental study
	5,6	-H ₆ + O		FTICR-MS	environmental	Curated database
9.984	1	-H ₂ + C		FTICR-MS	environmental	Experimental study
	5,6	-H ₂ + C		FTICR-MS	environmental	Curated database
10.021	1	-C ₂ H ₂ + O		FTICR-MS	environmental	Experimental study
	5,6	-C ₂ H ₂ + O		FTICR-MS	environmental	Curated database
11.964	1	-H ₄ + O		FTICR-MS	environmental	Experimental study
	5,6	-H ₄ + O		FTICR-MS	environmental	Curated database
12	1	C		FTICR-MS	environmental	Experimental study
	5	C		FTICR-MS	environmental	Curated database
12.036	1	-C ₂ H ₄ + O		FTICR-MS	environmental	Experimental study
	8	-C ₂ H ₄ + O		HRMS	living	Methods study
	5,6	-C ₂ H ₄ + O		FTICR-MS	environmental	Curated database
13.979	1	-H ₂ + O		FTICR-MS	environmental	Experimental study
	10	-H ₂ + O		LC-HRMS	environmental	Experimental study

	5,6	-H ₂ + O		FTICR-MS	environmental	Curated database
14.016	1	CH ₂		FTICR-MS	environmental	Experimental study
	9	CH ₂	methylation	FTICR-MS	environmental	Curated database
	5,6	CH ₂	methanol	FTICR-MS	environmental	Curated database
14.052	8	+C ₂ H ₆ - O		HRMS	living	Methods study
	5,6	+C ₂ H ₆ - O		FTICR-MS	environmental	Curated database
15.959	5	-C -H ₄ + O ₂		FTICR-MS	environmental	Curated database
15.995	1	O		FTICR-MS	environmental	Experimental study
	5,6,9	O	Oxidation/ Hydroxylation	FTICR-MS	environmental	Curated database
16.031	1	CH ₄		FTICR-MS	environmental	Experimental study
	5,6	CH ₄		FTICR-MS	environmental	Curated database
16.068	5,6	+C ₂ H ₈ - O		FTICR-MS	environmental	Curated database
17.974	5	-C - H ₂ + O ₂		FTICR-MS	environmental	Curated database
18.011	1	H ₂ O		FTICR-MS	environmental	Experimental study
	5,6,9	H ₂ O	Condensation/hydration/dehydration	FTICR-MS	environmental	Curated database
18.047	5,6	CH ₆		FTICR-MS	environmental	Curated database
19.99	5,6	-C + O ₂		FTICR-MS	environmental	Curated database
20.026	5,6	H ₄ O		FTICR-MS	environmental	Curated database
20.063	Not referenced					
21.984	5,6	+ C ₂ - H ₂		FTICR-MS	environmental	Curated database
23.964	5,6	+C - H ₄ + O		FTICR-MS	environmental	Curated database
24	5,6	C ₂		FTICR-MS	environmental	Curated database
24.036	5,6	+C ₃ H ₄ - O		FTICR-MS	environmental	Curated database

25.979	5,6	-H2 + C + O		FTICR-MS	environmental	Curated database
26.016	11	+O - C2H2		MS/MS	living	Review
	5,6,9	+O - C2H2		FTICR-MS	environmental	Curated database
26.052	5,6	C3H6O_1		FTICR-MS	environmental	Curated database
27.959	5,6	-H4 + O2		FTICR-MS	environmental	Curated database
27.995	5,6,9	CO	Formic Acid (-H2O)	FTICR-MS	environmental	Curated database
	12	CO		QTOF	living	Methods study
28.031	1	C2H4		FTICR-MS	environmental	Experimental study
	5,6,9	C2H4	Ethyl addition (-H2O)	FTICR-MS	environmental	Curated database
29.974	11	+O2 - CH2		MS/MS	living	Review
	5,6,9	+O2 - CH2	Nitro Reduction (-O2)			Curated database
30.011	11	CH2O		MS/MS	living	Review
	12	CH2O		QTOF	living	Experimental study
	5	CH2O		FTICR-MS	environmental	Curated database
30.047	13	C2H6	loss of 2 methyl radicals		living	Experimental study
	14	C2H6		QTOF	living	Methods study
31.99	11	O2		MS/MS	living	Review
	5,6	O2		FTICR-MS	environmental	Curated database
32.008	5,6	+ CH4 - O + S		FTICR-MS	environmental	Curated database
32.026	12	CH4O		QTOF	living	Methods study
	5,6	CH4O		FTICR-MS	environmental	Curated database
32.063	5,6	C2H8		FTICR-MS	environmental	Curated database
34.005	11	H2O2		MS/MS	living	Review
	5,6	H2O2		FTICR-MS	environmental	Curated database

34.042	5,6	CH6O		FTICR-MS	environmental	Curated database
35.964	Not referenced					
36	5,6	C3		FTICR-MS	environmental	Curated database
36.036	5	+ C4H4 - O		FTICR-MS	environmental	Curated database
36.058	Not referenced					
37.979	5,6	+ C2 - H2 + O		FTICR-MS	environmental	Curated database
38.016	5,6	C3H2		FTICR-MS	environmental	Curated database
39.959	5	+C - H4 + O2		FTICR-MS	environmental	Curated database
39.995	5	C2O		FTICR-MS	environmental	Curated database
40.031	5,6	C3H4		FTICR-MS	environmental	Curated database
41.974	5,6	C-H2+O2		FTICR-MS	environmental	Curated database
42.011	11	C2H2O		MS/MS	living	Review
	5,6,9	C2H2O	Acetylation (-H2)	FTICR-MS	environmental	Curated database
42.047	5,6	C3H6		FTICR-MS	environmental	Curated database
43.99	12	CO2		QTOF	living	Methods study
	5,6,9	CO2	Carboxylation	FTICR-MS	environmental	Curated database
44.008	Not referenced					
44.026	15	C2H4O	PEG / ethoxy repeating unit contamination	LC-HRMS	living	Methods study
	16	C2H4O	Ethoxy / ethylene oxide repeating unit (EO unit)	MALDI spiral-TOFMS	industrial	Experimental study
	5,6	C2H4O		FTICR-MS	environmental	Curated database
44.063	5,6	C3H8		FTICR-MS	environmental	Curated database

46.005	^{5,6}	CH2O2		FTICR-MS	environmental	Curated database
46.042	^{5,6}	C2H6O1		FTICR-MS	environmental	Curated database
48.021	^{5,6}	CH4O2		FTICR-MS	environmental	Curated database
48.058	^{5,6}	C2H8O		FTICR-MS	environmental	Curated database
51.995	^{5,6}	C3O		FTICR-MS	environmental	Curated database
52.031	^{5,6}	C4H4		FTICR-MS	environmental	Curated database
53.974	⁵	+ C2 - H2 + O2		FTICR-MS	environmental	Curated database
54.011	⁵	C3H2O		FTICR-MS	environmental	Curated database
54.047	⁵	C4H6		FTICR-MS	environmental	Curated database
55.99	^{5,9}	C2O2	Glyoxylate (-H2O)	FTICR-MS	environmental	Curated database
56.008	<i>Not referenced</i>					
56.026	^{5,6}	C3H4O		FTICR-MS	environmental	Curated database
56.063	^{5,6}	C4H8		FTICR-MS	environmental	Curated database
58.005	^{5,6}	C2H2O2		FTICR-MS	environmental	Curated database
58.042	^{5,6}	C3H6O		FTICR-MS	environmental	Curated database
60.021	^{5,6}	C2H4O2		FTICR-MS	environmental	Curated database
60.058	^{5,6}	C3H8O		FTICR-MS	environmental	Curated database
62.037	^{5,6}	C2H6O2		FTICR-MS	environmental	Curated database
66.011	^{5,6}	C4H2O		FTICR-MS	environmental	Curated database
67.99	^{5,6}	C3O2		FTICR-MS	environmental	Curated database
68.026	^{5,6}	C4H4O		FTICR-MS	environmental	Curated database
68.063	^{5,6}	C5H8		FTICR-MS	environmental	Curated database
70.005	^{5,6}	C3H2O2		FTICR-MS	environmental	Curated database
70.042	^{5,6}	C4H6O		FTICR-MS	environmental	Curated database

71.985	5,6	C2O3		FTICR-MS	environmental	Curated database
72.021	5,6	C3H4O2		FTICR-MS	environmental	Curated database
72.058	5,6	C4H8O		FTICR-MS	environmental	Curated database
74	6	C2H2O3		FTICR-MS	environmental	Curated database
74.037	5,6	C3H6O2		FTICR-MS	environmental	Curated database
76.052	<i>Not referenced</i>					
80.026	5,6	C5H4O		FTICR-MS	environmental	Curated database
82.005	5	C4H2O2		FTICR-MS	environmental	Curated database
82.042	5,6	C5H6O		FTICR-MS	environmental	Curated database
83.985	5,6	C3O3		FTICR-MS	environmental	Curated database
84.021	5,6,9	C4H4O2	Acetotacetate (-H2O)	FTICR-MS	environmental	Curated database
84.058	5,6	C5H8O		FTICR-MS	environmental	Curated database
86	6,9	C3H2O3	Malonyl_group_(-H2)	FTICR-MS	environmental	Curated database
86.037	5,6	C4H6O2		FTICR-MS	environmental	Curated database
88.016	5,6	C3H4O3		FTICR-MS	environmental	Curated database
88.052	5,6	C4H8O2		FTICR-MS	environmental	Curated database
94.005	5,6	C5H2O2		FTICR-MS	environmental	Curated database
96.021	5,6	C5H4O2		FTICR-MS	environmental	Curated database
98	6	C4H2O3		FTICR-MS	environmental	Curated database
98.037	5,6	C5H6O2		FTICR-MS	environmental	Curated database
100.016	5,6	C4H4O3		FTICR-MS	environmental	Curated database
100.052	5,6	C5H8O2		FTICR-MS	environmental	Curated database
102.032	5,9	C4H8O4	Erythrose (-H2O)	FTICR-MS	environmental	Curated database
110	6	C5H2O3		FTICR-MS	environmental	Curated database

112.052	5,6	C6H8O2		FTICR-MS	environmental	Curated database
114.032	5,6	C5H6O3		FTICR-MS	environmental	Curated database

- (1) Wang, L.; Lin, Y.; Li, J.; Yu, Q.; Xu, K.; Ren, H.; Geng, J. Deciphering Microbe-Mediated Dissolved Organic Matter Reactome in Wastewater Treatment Plants Using Directed Paired Mass Distance. *Environ. Sci. Technol.* **2024**, *58* (1), 739–750. <https://doi.org/10.1021/acs.est.3c06871>.
- (2) Linne, B. M.; Tello, E.; Simons, C. T.; Peterson, D. G. Chemical Characterization and Sensory Evaluation of a Phenolic-Rich Melanoidin Isolate Contributing to Coffee Astringency. *Food Funct.* **2025**, *16* (7), 2870–2880. <https://doi.org/10.1039/D4FO04934A>.
- (3) Lobodin, V. V.; Nyadong, L.; Ruddy, B. M.; Curtis, M.; Jones, P. R.; Rodgers, R. P.; Marshall, A. G. DART Fourier Transform Ion Cyclotron Resonance Mass Spectrometry for Analysis of Complex Organic Mixtures. *International Journal of Mass Spectrometry* **2015**, *378*, 186–192. <https://doi.org/10.1016/j.ijms.2014.07.050>.
- (4) Urban, J.; Jin, C.; Thomsson, K. A.; Karlsson, N. G.; Ives, C. M.; Fadda, E.; Bojar, D. Predicting Glycan Structure from Tandem Mass Spectrometry via Deep Learning. *Nat Methods* **2024**, *21* (7), 1206–1215. <https://doi.org/10.1038/s41592-024-02314-6>.
- (5) Danczak, R. E.; Goldman, A. E.; Chu, R. K.; Toyoda, J. G.; Garayburu-Caruso, V. A.; Tolić, N.; Graham, E. B.; Morad, J. W.; Renteria, L.; Wells, J. R.; Herzog, S. P.; Ward, A. S.; Stegen, J. C. Ecological Theory Applied to Environmental Metabolomes Reveals Compositional Divergence despite Conserved Molecular Properties. *Science of The Total Environment* **2021**, *788*, 147409. <https://doi.org/10.1016/j.scitotenv.2021.147409>.
- (6) Wu, M.; Li, P.; Li, G.; Liu, K.; Gao, G.; Ma, S.; Qiu, C.; Li, Z. Using Potential Molecular Transformation To Understand the Molecular Trade-Offs in Soil Dissolved Organic Matter. *Environ. Sci. Technol.* **2022**, *56* (16), 11827–11834. <https://doi.org/10.1021/acs.est.2c01137>.
- (7) Naim, A.; Hubert-Roux, M.; Cirriez, V.; Welle, A.; Vantomme, A.; Kirillov, E.; Carpentier, J.-F.; Giusti, P.; Afonso, C. Characterization of Modified Methylaluminoxane by Ion Mobility Spectrometry Mass Spectrometry and Ultra-High Resolution Fourier-Transform Ion Cyclotron Resonance Mass Spectrometry. *New J. Chem.* **2023**, *47* (46), 21244–21252. <https://doi.org/10.1039/D3NJ03877G>.
- (8) Yu, L.; Xiong, Y.-M.; Polfer, N. C. Periodicity of Monoisotopic Mass Isomers and Isobars in Proteomics. *Anal. Chem.* **2011**, *83* (20), 8019–8023. <https://doi.org/10.1021/ac201624t>.
- (9) Ayala-Ortiz, C.; Graf-Grachet, N.; Freire-Zapata, V.; Fudyma, J.; Hildebrand, G.; AminiTabrizi, R.; Howard-Varona, C.; Corilo, Y. E.; Hess, N.; Duhaime, M. B.; Sullivan, M. B.; Tfaily, M. M. MetaboDirect: An Analytical Pipeline for the Processing of FT-ICR MS-Based Metabolomic Data. *Microbiome* **2023**, *11* (1), 28. <https://doi.org/10.1186/s40168-023-01476-3>.
- (10) Oberleitner, D.; Schmid, R.; Schulz, W.; Bergmann, A.; Achten, C. Feature-Based Molecular Networking for Identification of Organic Micropollutants Including Metabolites by Non-Target Analysis Applied to Riverbank Filtration. *Anal Bioanal Chem* **2021**, *413* (21), 5291–5300. <https://doi.org/10.1007/s00216-021-03500-7>.
- (11) Gicquel, T.; Pelletier, R.; Bourdais, A.; Ferron, P.-J.; Morel, I.; Allard, P.-M.; Le Daré, B. Interest of Molecular Networking in Fundamental, Clinical and Forensic Toxicology: A State-of-the-Art Review. *TrAC Trends in Analytical Chemistry* **2024**, *172*, 117547. <https://doi.org/10.1016/j.trac.2024.117547>.
- (12) Ma, Y.; Kind, T.; Yang, D.; Leon, C.; Fiehn, O. MS2Analyzer: A Software for Small Molecule Substructure Annotations from Accurate Tandem Mass Spectra. *Anal. Chem.* **2014**, *86* (21), 10724–10731. <https://doi.org/10.1021/ac502818e>.

- (13) Cheng, C.; Yang, M.; Yu, K.; Guan, S.; Tao, S.; Millar, A.; Pang, X.; Guo, D. Identification of Metabolites of Ganoderic Acid D by Ultra-Performance Liquid Chromatography/Quadrupole Time-of-Flight Mass Spectrometry. *Drug Metabolism and Disposition* **2012**, 40 (12), 2307–2314. <https://doi.org/10.1124/dmd.112.047506>.
- (14) Aron, A.; Gentry, E.; Mcphail, K.; Nothias, L.-F.; Nothias-Esposito, M.; Bouslimani, A.; Petras, D.; Gauglitz, J.; Sikora, N.; Vargas, F. Reproducible Molecular Networking of Untargeted Mass Spectrometry Data Using GNPS.
- (15) Hess, S. *Sample Preparation Guide for Mass Spectrometry–Based Proteomics* | LCGC International. <https://www.chromatographyonline.com/view/sample-preparation-guide-mass-spectrometry-based-proteomics-0> (accessed 2026-04-23).
- (16) Sato, H.; Nakamura, S.; Teramoto, K.; Sato, T. Structural Characterization of Polymers by MALDI Spiral-TOF Mass Spectrometry Combined with Kendrick Mass Defect Analysis. *J. Am. Soc. Mass Spectrom.* **2014**, 25 (8), 1346–1355. <https://doi.org/10.1007/s13361-014-0915-y>.